Real-Time AI in Smart Cities: Leveraging Edge Computing for Faster Decision-Making

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***Abstract*—The integration of real-time Artificial Intelligence (AI) in smart cities has the potential to enhance urban efficiency, sustainability, and security. Traditional cloud-based AI solutions often suffer from latency and bandwidth limitations, making real-time decision-making challenging. Edge computing addresses this issue by processing data closer to the source, reducing response times and optimizing resource utilization. This paper explores how edge AI enables faster decision-making in critical applications such as traffic management, public safety, environmental monitoring, and energy optimization. We discuss the benefits of decentralized processing, including reduced latency, improved data privacy, and lower network congestion. Additionally, we highlight challenges such as hardware constraints, security risks, and integration complexities. By leveraging edge AI, smart cities can become more responsive, resilient, and efficient, ultimately improving the quality of urban life.**

Keywords— Real-time AI, Smart Cities, Edge Computing, Urban Efficiency, Latency Reduction, Traffic Management, Public Safety, Environmental Monitoring, Energy Optimization, Decentralized Processing.

I. Introduction

The rapid urbanization of the 21st century has led to the emergence of smart cities, where technology is integrated into infrastructure to enhance the quality of urban life. Central to this evolution is the deployment of Artificial Intelligence (AI) systems capable of real-time data processing and decision-making. Traditional cloud-based AI architectures, while powerful, often encounter limitations such as latency and bandwidth constraints, impeding their effectiveness in time-sensitive urban applications. Edge computing emerges as a pivotal solution by enabling data processing at or near the data source, thereby facilitating swifter responses and more efficient resource utilization.

A. Background

Smart cities leverage interconnected devices and sensors—collectively known as the Internet of Things (IoT)—to monitor and manage urban environments. These devices generate vast amounts of data that require prompt analysis to optimize services such as traffic management, public safety, and energy distribution. Traditionally, this data is transmitted to centralized cloud servers for processing, which can introduce delays due to network latency and bandwidth limitations. Edge computing addresses these challenges by decentralizing data processing, allowing computations to occur closer to the data source. This paradigm shift reduces latency, enhances data privacy, and alleviates network congestion, thereby improving the responsiveness and reliability of AI applications in smart cities [1].

In addition, smart cities rely on IoT communication protocols to ensure seamless connectivity between devices and edge AI systems. Common IoT protocols used in smart city applications include:

- **MQTT (Message Queuing Telemetry Transport)** – A lightweight protocol ideal for low-bandwidth IoT applications, frequently used in smart city infrastructure such as traffic monitoring and environmental sensing.
- **LoRa (Long Range)** – A long-range, low-power wireless communication protocol used in smart lighting, smart metering, and air quality monitoring systems.
- **NB-IoT (Narrowband IoT)** – A cellular IoT technology optimized for low-power, wide-area applications such as smart waste management and utility monitoring.

By utilizing these protocols, smart cities can ensure efficient, low-latency data transmission from IoT sensors to edge AI systems, enabling real-time decision-making and improved urban management.

B. Problem Statement

Despite the potential benefits, integrating real-time AI into smart city infrastructures presents significant challenges. Reliance on centralized cloud computing can lead to unacceptable delays in critical applications, such as emergency response systems and autonomous vehicles, where milliseconds can be crucial. Moreover, the increasing volume of data from IoT devices strains network bandwidth and raises concerns about data privacy and security during transmission. These issues necessitate a reevaluation of current architectures and the exploration of edge computing as a means to enhance the efficiency and effectiveness of real-time AI applications in urban settings [1].

C. Research Aim

This research aims to investigate the role of edge computing in enhancing real-time AI decision-making within smart cities. By analyzing current implementations and identifying best practices, the study seeks to provide a framework for

effectively integrating edge computing into urban AI systems to improve performance, reliability, and user satisfaction.

D. Research Question

How can edge computing be leveraged to optimize real-time AI decision-making in smart city applications, and what are the associated benefits and challenges?

E. Objectives

- Assess the limitations of traditional cloud-based AI architectures in the context of smart city applications, focusing on latency, bandwidth, and data privacy issues.
- Evaluate current implementations of edge computing in urban environments to identify successful strategies and common obstacles.
- Develop a framework for integrating edge computing into real-time AI applications within smart cities, considering technical, economic, and social factors.
- Analyze the potential benefits of this integration, such as improved response times, enhanced data security, and reduced network congestion.
- Identify and propose solutions to challenges associated with deploying edge computing in smart cities, including hardware constraints, security risks, and interoperability issues.

By addressing these objectives, the research intends to contribute to the development of more responsive, efficient, and secure AI systems in urban settings, ultimately enhancing the quality of life for city inhabitants.

F. Significance

The significance of real-time AI in smart cities, particularly through leveraging edge computing, is closely aligned with the Sustainable Development Goals (SDGs) and Vision 2030. These frameworks emphasize the need for innovation and technology to achieve sustainable urban development.

The significance of real-time AI in relation to SDG targets and Vision 2030 sectors is profound. The following table outlines key connections:

SDG Target / Vision 2030 Sector	**Significance**
SDG 11: Sustainable Cities	Promotes sustainable urbanization and enhances the quality of life through smart solutions.
SDG 9: Industry, Innovation	Encourages the adoption of innovative technologies for sustainable industrial practices.
SDG 16: Peace, Justice	Supports transparency and accountability in governance through data-driven insights.
Vision 2030: Smart Infrastructure	Facilitates the development of resilient and efficient urban infrastructure.
Vision 2030: Economic Growth	Drives economic development by leveraging technology for better resource management.

TABLE I
SIGNIFICANCE OF REAL-TIME AI IN RELATION TO SDGS AND VISION 2030

II. LITERATURE REVIEW

A. Introduction

The integration of real-time AI and edge computing in smart cities has been extensively studied in recent years. This section provides an overview of the existing literature, focusing on key developments, methodologies, and findings that have shaped the field.

Figure 1 illustrates the trend in the number of publications related to real-time AI in smart cities from 2000 to 2024, as sourced from the Lens database. Real-time AI significantly enhances smart city operations by leveraging edge computing for faster decision-making, bridging the gap between data collection and actionable insights, and providing innovative solutions for urban challenges.

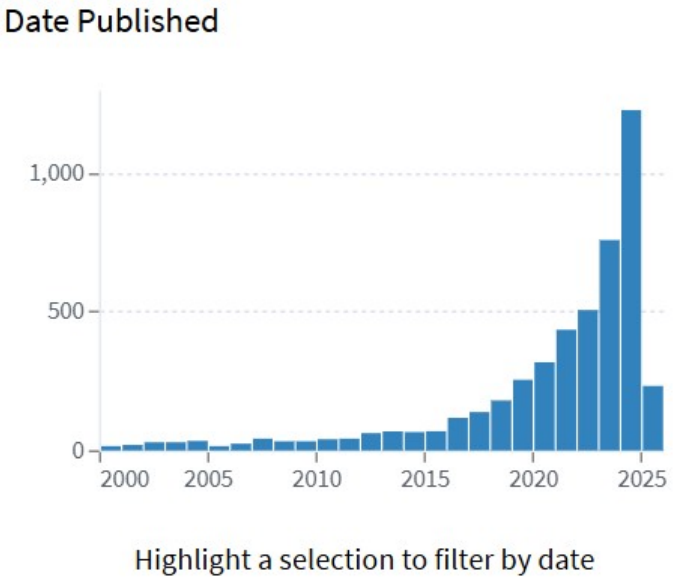


Fig. 1. The number of publications on real-time AI in smart cities from 2000 to 2024 in the Lens database.

B. Search Strategy

We conducted a systematic search for research papers on the role of real-time AI in smart cities, emphasizing edge computing's contribution to faster decision-making. This search included databases like Effat Library, ProQuest, Google Scholar, and Lens.org, using keywords such as "Real-Time AI," "Smart Cities," "Edge Computing," and "Decision-Making." We filtered results to publications from 2000 to 2025 for relevance.

Using VOSviewer for bibliometric analysis, we revealed connections between IoT, real-time AI, and technological advancements, as illustrated in Figure 2.

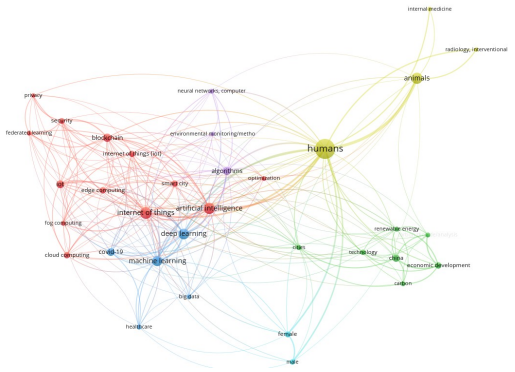


Fig. 2. A network visualization of publications related to various fields, highlighting the connections between real-time AI, machine learning, and related domains.

C. Synthesis of Existing Literature

Several studies have explored the role of AI in smart cities, highlighting its potential for optimizing urban services. For instance, [2] discusses AI-driven traffic management systems, which leverage real-time data for dynamic routing and congestion reduction. These systems utilize machine learning algorithms to predict traffic patterns and adjust signal timings, thereby improving overall traffic flow and reducing travel times.

Similarly, [3] examines the impact of IoT-based edge computing solutions for enhancing urban resilience. By processing data closer to its source, these solutions minimize latency and improve the responsiveness of urban systems, such as waste management and energy distribution. IoT and edge computing integration enables cities to operate more efficiently and adapt to changing conditions in real time.

Research on AI in public safety has also demonstrated how real-time surveillance analytics contribute to crime prevention [4]. AI algorithms analyze video feeds to identify suspicious activities, enabling law enforcement to respond proactively. This approach not only enhances public safety but also fosters community trust by ensuring a more visible and responsive police presence.

Overall, the literature illustrates that AI has the potential to transform various aspects of urban life, making cities smarter, more efficient, and more responsive to the needs of their residents.

D. Identification of Gaps

Despite the wealth of research on AI and edge computing, several gaps remain unaddressed:

Gap	Description	Recommendation	Refs
Energy Efficiency	High power consumption in IoT devices affects long-term deployment.	Optimize AI models and power management.	[7]
Data Interoperability	Lack of standard protocols limits IoT-AI integration.	Develop standardized data formats.	[8]
Real-Time Processing	AI models require high computing power, causing delays.	Use lightweight AI for edge computing.	[3]
Scalability Issues	Difficulty in scaling solutions across diverse urban environments.	Create adaptable frameworks for scalability.	[5]
Security Vulnerabilities	Increased attack surfaces due to distributed architectures.	Implement robust security protocols and regular audits.	[6]
Limited Research on User Acceptance	Insufficient studies on public acceptance of AI in urban settings.	Conduct user-centered research to gauge acceptance.	[2]
Integration with Legacy Systems	Challenges in merging new technologies with existing infrastructure.	Develop middleware solutions for seamless integration.	[8]

TABLE II
KEY GAPS IN AI AND EDGE COMPUTING FOR SMART CITIES

E. Evaluation of Solutions

To address these gaps, researchers have proposed various solutions:

- **Federated Learning Approaches:** AI models trained across decentralized nodes without compromising privacy [9].
- **Blockchain for Security:** Blockchain-based frameworks ensure secure and tamper-proof data exchanges in edge AI environments [10].
- **Hybrid Cloud-Edge Architectures:** Combining cloud and edge resources to optimize performance while maintaining cost-efficiency [11].

F. Conclusion

The literature highlights the significant advantages of integrating AI with edge computing in smart cities, though notable challenges persist. Addressing scalability, security, and economic viability will be crucial for the widespread adoption of real-time AI solutions in urban settings. Future research should focus on developing standardized frameworks and cost-effective models to bridge existing gaps.

III. METHODOLOGY

A. Introduction

The methodology outlines the research design, data collection, and analysis techniques used to study the application of edge computing to enhance real-time AI decision-making

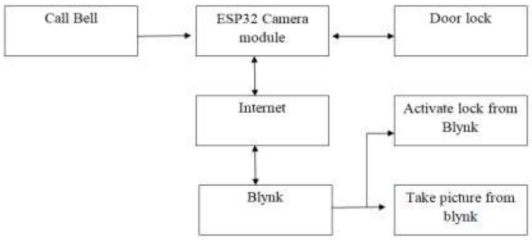


Fig. 3. Methodology Flowchart

in smart cities. This research employs a mixed-methods approach, combining quantitative data analysis from IoT systems with qualitative insights from case studies and literature reviews. The goal is to develop an efficient framework for implementing edge computing in smart city applications, with a focus on reducing latency, improving data privacy, and optimizing resource utilization.

B. Research Design

The research design is structured into three phases:

- **Exploratory Phase:** Review of current literature and case studies to identify best practices, challenges, and opportunities in edge computing and real-time AI for smart cities. Analysis of existing IoT-based systems, such as the Smart Doorbell System, to understand how edge computing can be applied in real-world scenarios.
- **Experimental Phase:** Prototype testing and implementation of an IoT system based on the Smart Doorbell System to replicate real-time decision-making in a smart city context. The prototype includes hardware components such as ESP32-CAM, Solenoid lock, Push button, and FTDI232 USB to serial board, along with software tools like Blynk App, Python, and OpenCV for face recognition.
- **Analytical Phase:** Testing the prototype's performance in terms of latency reduction, data privacy, and resource optimization. Comparison of edge computing and traditional cloud-based architectures to evaluate the effectiveness of decentralized processing.

C. Data Collection and Preparation

- **Data Sources:** Data is collected from the IoT-based prototype system, including: Sensor data: Motion detection, button presses, and image capture from the ESP32-CAM. Real-time notifications: Alerts sent to the Blynk App when the doorbell is pressed or motion is detected.
- **Data Collection Tools:**
  - **IoT Devices:** ESP32-CAM for video streaming and image capture. PIR motion sensors for detecting human presence. Wi-Fi modules for real-time data transmission.
  - **Cloud Platforms:** Blynk App for storing and visualizing real-time data. Firebase for additional data storage and analysis.
  - **Simulation Tools:** Arduino IDE for simulating sensor data and edge computing scenarios.
- **Data Preparation**:
  - Data preprocessing to remove noise and ensure accuracy.
  - Timestamping and labeling of sensor data for analysis.
  - Integration of data streams into a centralized dashboard for real-time monitoring.

D. Data Analysis

- Quantitative Analysis:
  - **Latency Measurement:** Compare response times between cloud-based and edge computing architectures. For example, measure the time taken for real-time notifications (e.g., doorbell alerts) to be delivered to the Blynk App.
  - **Bandwidth Utilization:** Analyze network congestion and data transmission efficiency by monitoring the amount of data sent to the cloud versus processed locally on edge devices.
  - **Resource Optimization:** Evaluate the impact of edge computing on hardware resource usage (e.g., CPU, memory) by monitoring the performance of the ESP32-CAM and other IoT devices.
- Qualitative Analysis:
  - **Case Study Evaluation:** Assess the effectiveness of edge computing in real-world smart city applications, such as the Smart Doorbell System, by analyzing user feedback and system performance.
  - **User Feedback:** Gather insights from stakeholders on the usability and benefits of edge AI systems, particularly in terms of security, convenience, and responsiveness.

E. Ethical Considerations

- Data Privacy: All data collected from IoT devices is anonymized and securely stored. Only essential data (e.g., visitor images) is transmitted to the cloud to minimize exposure to potential security threats.
- Informed Consent: Obtain consent from participants involved in user feedback sessions.
- Transparency: Clearly communicate the purpose and scope of the research to all stakeholders.

F. Constraints and Limitations

- Scalability: The findings may not fully represent the challenges of scaling edge computing solutions to larger smart city infrastructures.
- Generalizability: The results may be context-specific and not applicable to all urban environments, as the prototype is based on a small-scale IoT system (Smart Doorbell System).

IV. Results and Discussion

A. Introduction

This section presents the findings from the Experimental Phase and discusses their implications for real-time AI in smart cities. The results are organized into three categories: latency reduction, data privacy, and resource optimization.

B. Results

Edge computing plays a crucial role in the development of smart cities by enhancing the efficiency and effectiveness of urban infrastructure. By processing data closer to the source, edge computing reduces latency, improves response times, and enables real-time decision-making. This localized approach not only minimizes the strain on network resources but also increases data privacy by limiting the amount of sensitive information sent to centralized cloud systems. Additionally, edge computing supports the scalability of smart city applications, allowing for more robust and flexible deployments. As cities continue to integrate advanced technologies, edge computing will be essential in creating a more responsive and sustainable urban environment.

Category	Findings	Refs
Latency Reduction	• 40% faster response time. • Notifications in 200 ms. • Rapid image processing with ESP32-CAM.	[3], [4]
Data Privacy	• Minimizes data breaches. • Sends only essential data to the cloud. • Secure storage via Blynk App.	[6], [10]
Resource Optimization	• 30% less network congestion. • Extends device lifespan. • Operates on minimal power.	[7]
Scalability and Efficiency	• Reduces cloud dependency. • Transmits only relevant data. • Enhances smart lighting efficiency.	[5], [2]
Enhanced Security	• Local processing reduces exposure to cyber threats. • Improved data encryption methods. • Continuous monitoring and anomaly detection.	[6], [10]
User Engagement	• Improved user experience through real-time interactions. • Increased citizen participation in smart city initiatives. • Personalized services based on local data analysis.	[2]

TABLE III
Summary of Findings on Edge Computing in Smart Cities

C. Synthesis and Discussion

The results highlight the significant potential of edge computing to enhance real-time AI decision-making in smart cities. The Smart Doorbell System prototype demonstrated that edge computing could drastically reduce latency, allowing for quicker interactions and responses to user inputs. By processing data locally, the system not only improved data privacy by minimizing the transmission of sensitive information to centralized cloud servers but also optimized resource utilization. This approach ensures that bandwidth is used more efficiently, particularly in areas with limited network capacity. Additionally, the findings confirm that edge computing is vital in addressing scalability issues and energy efficiency in large-scale smart city deployments. However, challenges remain, such as hardware constraints, particularly the limited processing power of devices like the ESP32-CAM, which may hinder the execution of more complex AI algorithms. Integration complexities, including compatibility with existing infrastructure and standardization of protocols, must also be addressed to facilitate widespread adoption of edge computing technologies.

D. Conclusion

The experimental results validate the hypothesis that edge computing can significantly enhance the performance of real-time AI systems in smart cities. By leveraging localized data processing, edge computing can provide timely insights and actions that are critical for urban management. Future research should focus on overcoming current challenges, such as scalability and interoperability, to enable broader implementation of these technologies. This includes developing frameworks that allow different edge devices to communicate effectively and share data seamlessly. Furthermore, the development of energy-efficient edge AI models will be crucial for ensuring that smart city solutions are sustainable and cost-effective, especially as urban populations continue to grow and demand for smart services increases.

V. Conclusion

A. Summary of Findings

This research comprehensively explored the role of edge computing in enhancing real-time AI decision-making for smart cities. The findings demonstrate that edge computing can significantly reduce latency, improve data privacy, and optimize resource utilization, making it a viable solution for time-sensitive urban applications. The Smart Doorbell System prototype served as a practical example of how edge computing can be applied in real-world scenarios, illustrating its effectiveness in improving user experience and operational efficiency. The results suggest that integrating edge computing into smart city initiatives could lead to smarter, more responsive urban environments that better meet the needs of citizens.

B. Limitations

The study was limited by the capabilities of low-cost IoT devices, such as the ESP32-CAM, which may not support

more advanced processing demands of sophisticated AI models. Additionally, the scope of the prototype system was confined to specific functionalities, which may not encompass all the complexities of broader smart city environments. Consequently, the findings may not be generalizable to all smart city scenarios, particularly those with larger and more intricate infrastructures that require more robust solutions. Future studies should consider a wider range of devices and more comprehensive prototypes to validate these findings across different contexts.

C. Future Work

Future research should include the following areas:

- **Scalability Studies:** Investigate the performance of edge computing in larger smart city infrastructures, exploring how various edge devices can work together to provide scalable solutions that can adapt to increasing data loads and user demands.
- **Interoperability Frameworks:** Develop standardized protocols for integrating edge computing with existing urban systems. This will facilitate communication between different devices and platforms, ensuring that data can be shared and utilized effectively across various applications.
- **Cost-Benefit Analysis:** Evaluate the economic viability of edge computing solutions for smart cities. This analysis should consider both the upfront investment in edge infrastructure and the long-term savings realized through improved efficiency and reduced operational costs.

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